Unconstrained Feature Enhancement Text-Guided Few-Shot Remote Sensing Image Object Detector

Xiping Shang, Wei Zhao, Haoxiang Chen, Xudong Fan, Nannan Li, Dongjin Li, Jianwei Lv, and Rufei Zhang

***Abstract*—Few-shot object detection (FSOD) for remote sensing images (RSIs) has received much attention. The main difficulty in FSOD currently lies in the imprecise classification, which can be tackled from three key perspectives: features to be classified (region of interest (RoI) features), reference features (feature prototypes), and classifiers. However, existing methodologies have not comprehensively addressed these aspects. To fill this gap, we propose the unconstrained feature enhancement text-guided few-shot remote sensing image object detector (UFEDet). Initially, we introduce the text–image shared-specific module (SSM), which not only focuses on the consistent representation of the same class across different modalities but also emphasizes the unique information of each modality, thereby enhancing the distinctiveness of class feature prototypes. Subsequently, we propose an unconstrained feature enhancement module, which enhances the RoI features by leveraging the global class information from feature prototypes, while also increasing the distributional differences between classifiers. Finally, we introduce a gradient-controlled optimizer that modulates the gradient based on the gradient response of base classes, alleviating catastrophic forgetting of base classes during the training of novel classes. To validate the efficacy of our proposed approach, experiments were conducted on the DIOR and NWPU VHR-10 datasets. The experimental results demonstrate that the mAP of both base and novel classes exceeds that of popular few-shot object detectors.**

***Index Terms*—Few-shot learning, object detection, remote sensing images (RSIs), text-modal knowledge.**

I. Introduction

In RECENT years, object detection networks based on deep convolutional neural networks have made significant advancements in the field of remote sensing imagery [1], [2], [3]. However, data-driven CNN-based object detection frameworks are prone to overfitting when training samples are limited. Therefore, object detection in remote sensing imagery with a small number of training samples is a significant and challenging task [4].

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Xiping Shang, Wei Zhao, Haoxiang Chen, and Xudong Fan are with the School of Electronic and Information Engineering, Beihang University, Beijing 100191, China (e-mail: BY2102133@buaa.edu.cn; zhaowei203@buaa.edu.cn; hxchen1230@buaa.edu.cn; XudongFAN@buaa.edu.cn).

Nannan Li, Dongjin Li, Jianwei Lv, and Rufei Zhang are with Beijing Institute of Control and Electronics Technology, Beijing 100038, China (e-mail: beyondlnn@163.com; dongjinddj@163.com; 1637804619@qq.com; zrf_lucky@sina.com).

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Currently, there are two main learning architectures for few-shot object detection (FSOD). The first is based on fine-tuning methods [5], [6], [7], [8], which accomplish the task of FSOD by fine-tuning the weights of a large-sample-driven base class detector on a novel class dataset. The second is based on meta-learning methods [9], [10], [11], [12], which construct a series of sub-tasks (episodes) from the base classes and novel classes during the training stage. The FSOD methods based on fine-tuning essentially continue the learning approach of general detectors, making it difficult to break through the constraints of a few samples and learn the generalization features of novel classes. Therefore, in subsequent research on FSOD methods, the classic meta-learning method Meta R-CNN is chosen as the baseline for improvement.

Upon further analysis of the performance of FSOD methods, the primary errors for novel classes are concentrated in the classification branch rather than the regression branch. The FSOD methods effectively identify foreground proposals, but the model misclassifies novel classes as either confusable base classes or background [13], as shown in Fig. 1. The root causes of classification errors lie in three aspects.

1) The differences between the region of interest (RoI) features of novel classes and those of base classes have not been successfully captured [15], [16]. In object detection tasks, RoI features serve as critical information for distinguishing objects. However, in a few-shot learning scenario, the limited number of samples for novel classes prevents models from adequately learning their unique visual features. This leads to significant overlap between the RoI features of the novel and base classes, making effective differentiation challenging. To solve this problem, researchers have emphasized the importance of feature prototypes in enhancing the discriminability of RoI features [9], [17].
2) However, under a few-shot condition, feature prototypes struggle to capture subtle differences between classes. A feature prototype is a representative feature vector for a class, constructed from limited samples. With scarce training data (e.g., 5–10 images), these prototypes do not adequately capture all variations in a class, particularly subtle distinctions. This limitation results in imprecise representations of novel classes, increasing their likelihood of being misclassified as base classes with similar prototypes. Although researchers have explored methods to improve prototype discriminability [12], [16], [18], prototypes derived solely from image data are

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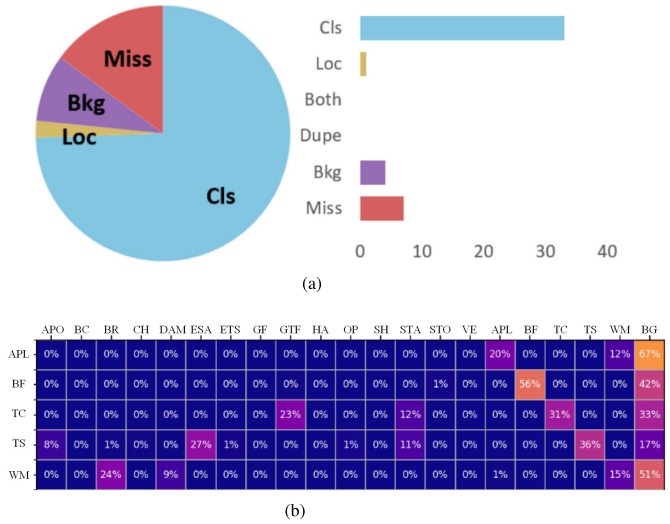


Fig. 1. (a) False analysis on DIOR dataset with 10-shot setting [13]. Using the toolbox for identifying detection errors (TIDEs) [13], [14] and calculating the confusion matrix, the primary error type is classification errors, which misclassify novel classes as either confusable base classes (Cls) or backgrounds (Miss). (b) Object classes of confusion matrix include airplane (APL), airport (APO), baseballfield (BF), basketball-court (BC), bridge (BR), chimney (CH), expressway-service-area (ESA), expressway-toll-station (ETS), dam (DAM), golf-field (GF), ground-track-field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage-tank (STO), tennis-court (TC), train-station (TS), vehicle (VE), windmill (WM) and background (BG).

prone to overfitting. To address this, some scholars integrate textual information [13] to enhance prototypes. However, these methods prioritize consistency within the same class across modalities (e.g., text–image alignment) while neglecting the unique characteristics of novel classes.

3) Due to insufficient training data, classifiers tend to misclassify novel classes into more frequent base classes. Practically, the scarcity of novel class samples limits a classifier's ability to adjust its parameters to match their feature distributions. Consequently, when encountering ambiguous samples, classifiers default to assigning them to base classes, which have more extensive training data (proof in Section III-D).

   In addition, most of the existing FSOD methods [20], [21], [22], [23], [24], [25] merely focus on the performance of novel classes while ignoring the degradation of the base classes, resulting in the catastrophic forgetting problem.

To solve the above problems, our research proposes the unconstrained feature enhancement text-guided few-shot remote sensing image object detector (UFEDet). The main contributions of this article are summarized as follows.

1) We have introduced the text–image shared-specific module (SSM), which partitions features into consistent and distinctive characteristics. By enforcing the maximization of correlation among consistent features across different modalities and the minimization of correlation among distinctive features, the module enhances the discriminability of class feature prototypes.
2) We propose an unconstrained feature enhancement module (UFEM), which enhances the RoI features by leveraging the global class information from feature prototypes, while also overcoming the issue of minimal distribution differences in classifiers due to the number of samples.
3) We introduce a gradient-controlled optimizer that modulates the gradient based on the gradient response of base classes, alleviating catastrophic forgetting of base classes during the training of novel classes.

The remaining paper is organized as follows. We briefly summarize the related works in Section II. We present the proposed framework in Section III and conducted extensive experiments in Section IV to validate its effectiveness. Section V concludes this work.

II. Related Work

In recent years, in the field of remote sensing imagery, work related to FSOD has attracted much attention. For methods based on fine-tuning, PAMS-Det [20] proposed a path-aggregation module that shortens the information transmission path through a top-down process and utilizes low-level semantic information for localization. Context information refinement (CIR)-FSD [21] designed a CIR module to extract discriminative context features to address the interference of complex backgrounds on FSOD. SAM& BFS [22] proposed a few-shot fine-tuning network with a shared attention module to adapt to the detection of remote sensing targets with significant size variations. TFACSC [23] proposed a multihead scaled cosine nonlocal model to reduce the occurrence of category confusion. ST-FSOD [24] proposed a self-training FSOD network that uses pseudo-labels to improve the recall rate of unlabeled new classes.

For meta-learning methods, FSODM [25] designed a weight adjustment module to optimize feature prototypes and enhance discriminability. SAGS-TFS [26] proposed a bidirectional foreground stimulation module, applying similar graphs simultaneously to feature prototypes and RoI features, increasing interclass variance and reducing intra-class variance. SAAN [27] uses adaptive attention to enhance target features rather than the entire image, avoiding interference from irrelevant information. P-CNN [28] proposed a prototype-guided region proposal network (P-G RPN) to identify foreground targets from complex backgrounds. MM-RCNN [29] employs a memory module to store the knowledge learned about each class during the training stage, and the memory-external attention (MEA) aggregates information from all classes simultaneously. Self-training-based region proposal networks (RPNs) [30], incorporating self-training bounding box head modules, have also been developed to improve proposal generation for novel object detection. SAE-FSDet [31] employs a gradual RPN to generate high-quality proposals, improving the recall of novel object detection, alongside a label-consistent classifier (LCC) to address label assignment inconsistencies. SMDC-Net [32] enhances feature representation by integrating salient object detection with a multihead detector. CCFR [33] uses contrastive learning and class-attention reweighting to boost novel-class feature discriminability in complex

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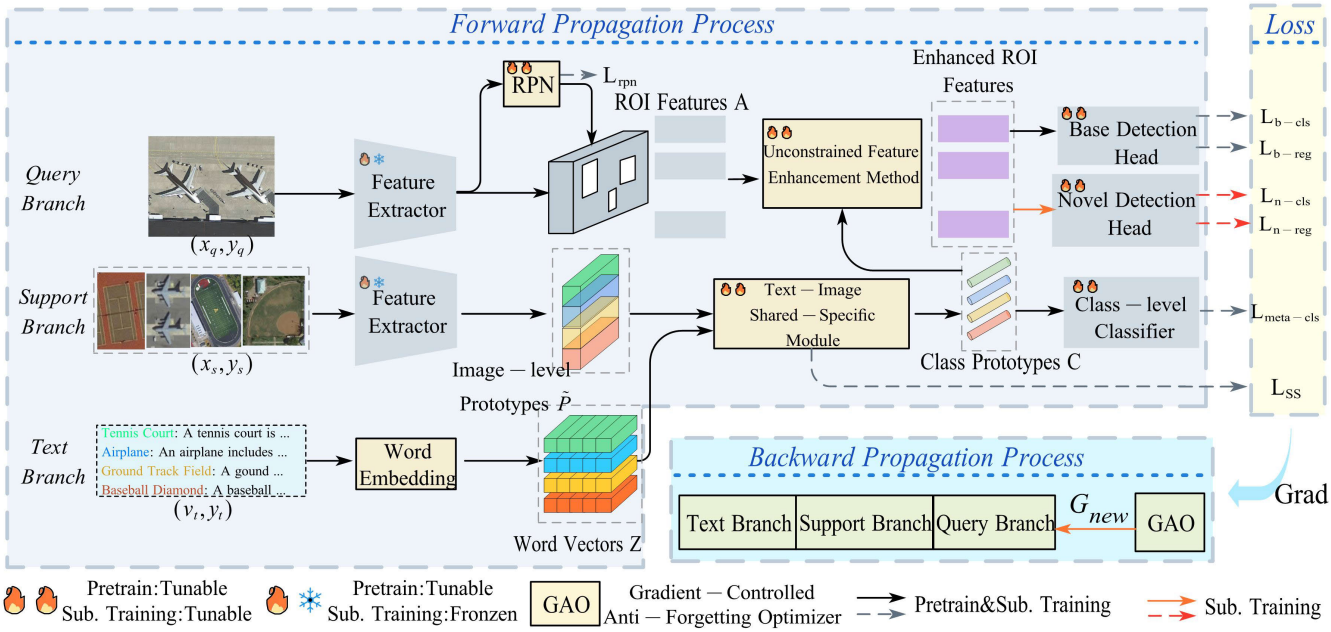


Fig. 2. Architecture of our UFEDet.

remote sensing scenes. MFKDIOD [34] combines multiscale knowledge distillation and implicit object discovery to tackle base-class forgetting and unlabeled instance interference in few-shot remote sensing detection. FSODS [35] proposes a few-shot detection framework extendable to SAR images, leveraging multiscale knowledge distillation to adapt to radar backscatter's spatial features and implicit object discovery to mitigate occlusion/speckle noise interference, enabling robust target detection under limited annotations.

Existing FSOD methods primarily focus on addressing classification errors through RoI features and feature prototypes, without adequately considering the importance of the distributional differences among classifier parameters.

III. Method

A. Problem Statement

FSOD in remote sensing images (RSIs) faces issues like weak novel-class feature prototype representation, novel-class classifier errors, and catastrophic forgetting. To address these, a text-guided unconstrained feature enhancement detection model is proposed. It enhances feature prototype distinguishability, strengthens RoI feature representation, and alleviates catastrophic forgetting of base classes during novel-class training.

In the problem definition of few-shot learning, the collection of all classes is denoted as Y. The classes with a large number of labeled samples are referred to as base classes, denoted as Y_b, while the classes with a limited number of labeled samples are called novel classes, denoted as Y_n. Y_b and Y_n satisfy $Y_b \cup Y_n = Y$ and $Y_b \cap Y_n = \emptyset$. In the process of few-shot learning, it is common to divide it into two stages: pretraining and subsequent training. The image spaces for base classes and novel classes are defined as X_b and X_n, respectively. During the pretraining stage, the model learns a transferable feature extraction model on the base class dataset $D_b = \{(x, y)|x \in X_b, y \in Y_b\}$ to obtain a robust and universal representation. In the subsequent training stage, the model uses the prior knowledge learned from the first stage to fine-tune using both base classes D_b and novel classes $D_n = \{(x, y)|x \in X_n, y \in Y_n\}$.

In particular, subsequent training involves randomly sampling data from D_b and D_n to form a series of N-way K-shot tasks. Each task includes N classes, with only K labeled samples provided for each class for subsequent training. These $N \times K$ samples constitute the training samples for subsequent training. D_b and D_n to be reconstructed into a series of episodes. Each episode consists of a query set Q and a support set S

$$\begin{aligned} Q &= \left\{\left(x_q, y_q\right) | x_q \in X_b \cup X_n, y_q \in Y_b \cup Y_n\right\} \\ S &= \left\{\left(x_s, y_s\right) | x_s \in X_b \cup X_n, y_s \in Y_b \cup Y_n\right\} \end{aligned} \tag{1}$$

where (x_q, y_q) represents the query image and its corresponding annotations. (x_s, y_s) represents the object instances cropped from the support image and their class labels.

Similarly, the text set can be represented as

$$T = \{(v_t, y_t) | v_t \in V_b \cup V_n, y_t \in Y_b \cup Y_n\} \tag{2}$$

where V_b and V_n represent the text set of the base class and the novel class, respectively.

B. Overview

As shown in Fig. 2, our proposed UFEDet mainly consists of the forward propagation, loss calculation, and backward propagation.

The forward propagation is composed of the query branch, support branch, and text branch. Given (x_s, y_s) and (v_t, y_t), the support branch and text branch, based on the extraction

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of image-level feature prototypes $\tilde{P}$ and word vectors $\tilde{Z}$, utilize the text–image SSM to maximize consistent features and minimize correlation while preserving the distinctiveness of different modalities. This process leads to the generation of more generalizable feature prototypes C. Given (x_q, y_q), the query branch, after extracting RoI features A, enhances the RoI features by introducing feature prototypes through the UFEM. It is important to note that the UFEM module does not adopt a sample-guided enhancement strategy but instead proposes a classifier-guided feature prototype enhancement strategy, which reduces the impact of sample imbalance on the classifier distribution.

During the backward propagation, the GAO is introduced. This module adjusts the overall gradient Grad based on the gradient changes of the base categories, thereby mitigating the impact of catastrophic forgetting.

Loss calculation is only used during the training process and has different compositions in the pretraining stage and the subsequent training stage. In the pretraining stage, the loss is

$$L_{\text{pretrain}} = L_{\text{rpn}} + L_{b-\text{cls}} + L_{b-\text{reg}} + L_{\text{meta}} + L_{\text{SS}} \tag{3}$$

where L_{rpn} represents the loss of RPN. $L_{b-\text{reg}}$ and $L_{b-\text{cls}}$ denote regression loss and classification loss of the base class, respectively. L_{meta} is proposed in Meta R-CNN. L_{SS} is the shared-specific loss. In the subsequent training stage, the loss is

$$L_{\text{subtrain}} = L_{\text{rpn}} + L_{b-\text{cls}} + L_{b-\text{reg}} + L_{n-\text{cls}} + L_{n-\text{reg}} + L_{\text{meta}} + L_{\text{SS}} \tag{4}$$

where $L_{n-\text{reg}}$ and $L_{n-\text{cls}}$ denote regression loss and classification loss of the novel class, respectively.

C. Text–Image SSM

RSIs, compared to natural images, contain more complex background information, greater variations in scales. Especially in scenarios with few samples, these issues become even more challenging. In such cases, combining visual information with textual information is an effective approach. However, existing methods primarily focus on the consistency among different modalities within the same class, while neglecting their uniqueness. To address this issue, the text–image SSM is proposed to mitigate the impact, as shown in Fig. 3.

First, extract features from the sentence description

$$\tilde{Z}_i = \text{Lstm}\,(M \times Z_i) \tag{5}$$

where M represents the random masking strategy [13], Z_i represents the word vectors of class i, Lstm represents the long short-term memory network, and $\tilde{Z}_i$ represents the word vectors after masking.

Given the text-level feature prototypes $\tilde{Z} \in R^{N\times d}$ and the image-level feature prototypes $\tilde{P} \in R^{N\times d}$. d represents the dimension. The shared-matrix J_c and the specific-matrix J_u are formulated as follows:

$$\begin{aligned} J_c &= \left(\tilde{Z}^1 W_2\right)\left(\tilde{P}^1 W_3\right)^{\text{T}} \\ J_u &= \left(\tilde{Z}^2 W_4\right)\left(\tilde{P}^2 W_5\right)^{\text{T}} \end{aligned} \tag{6}$$

where W_2, W_3, W_4, and $W_5 \in R^{d/2\times 1}$ represent weight matrices, the concatenation of common part $\tilde{Z}^1 \in R^{N\times d/2}$ and unique part $\tilde{Z}^2 \in R^{N\times d/2}$ results in $\tilde{Z}$, the concatenation of common part $\tilde{P}^1 \in R^{N\times d/2}$ and unique part $\tilde{P}^2 \in R^{N\times d/2}$ results in $\tilde{P}$. The relationship matrix quantifies how closely each feature prototype relates to others. For the J_c, we want the diagonal of this matrix (representing similarity) to approach 1, indicating strong consistency between text features and image features. For the J_u, the diagonal should approach 0, suggesting minimal similarity between text features and image features, thus emphasizing distinctiveness.

An image-text shared-specific loss is proposed. This loss function is designed to optimize the relationship matrix. It encourages the common components to be highly consistent (diagonal near 1) and the unique components to be distinct (diagonal near 0). This ensures that the model learns both shared and unique features effectively, as detailed as follows:

$$\begin{aligned} L_c &= \sum_{i}^{N}\left(1 - J_{c,ii}\right)^2 + \lambda_c \sum_{i}^{N}\sum_{j\neq i}^{N} J_{c,ij}^2 \\ L_u &= \sum_{i}^{N}\left(J_{u,ii}\right)^2 + \lambda_u \sum_{i}^{N}\sum_{j\neq i}^{N} J_{u,ij}^2 \\ L_{\text{SS}} &= L_c + L_u \end{aligned} \tag{7}$$

where L_{SS} represents the image-text shared-specific loss, L_c represents the image-text shared loss, L_u represents the image-text specific loss, λ_c and λ_u are positive constant.

To enhance the discriminability of feature prototypes, we fuse text-level feature prototypes $\tilde{Z}$ with image-level feature prototypes $\tilde{P}$, detailed as follows:

$$C = \tilde{Z} + \tilde{P} \tag{8}$$

where $C \in R^{N\times d}$ represents final class feature prototypes, N represents the number of classes.

D. Unconstrained Feature Enhancement Module

We propose an UFEM that reduces the impact of sample size on the classifiers during the process of enhancing classifiers using feature prototypes. This method not only enhances RoI features but also increases the distributional differences between classifiers, which can be presented as follows:

$$\tilde{S}_{\text{cls}} = \sum_{i=1}^{N} \text{AC}_i^* \hat{F}_i^{\text{T}} + \frac{1}{N}\sum_{i=1}^{N} \text{AC}_i^* \hat{F}_{N+1}^{T} \tag{9}$$

where $A \in R^{n\times d}$ represents RoI features. n represents the number of proposal boxes in the current image. $C_i \in R^{1\times d}$, $C_i = (C_{i,1}, C_{i,2}, \ldots, C_{i,d})$ represents the feature prototype for class i. $C_i^* = \text{diag}(C_{i,1}, C_{i,2}, \ldots, C_{i,d})$ represents the diagonal matrix of C_i. $\hat{F}_i \in R^{(N+1)\times d}$, $\hat{F}_i = \{0, 0, \ldots, \tilde{f}_i, \ldots, 0\}^{\text{T}}$, $N+1$ represents the number of classes add the background. $\tilde{f}_i \in R^{1\times d}$ represents the parameters for class i classifier. $\tilde{S}_{cls} \in R^{n\times(N+1)}$, $\tilde{S}_{cls} = (\tilde{s}_1, \tilde{s}_2, \ldots, \tilde{s}_{N+1})$ represents the output results of class classifiers. $\tilde{s}_i \in R^{n\times 1}$ represents the results from each class classifier.

Below is the proof that UFEM can increase the distributional differences between classifiers. We use cross-entropy for the

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classification loss ($L_{b-\mathrm{cls}}, L_{n-\mathrm{cls}}$) in the forward computation for training, as follows:

$$\mathrm{Loss} = -\sum_{i=1}^{N+1} y_i \ln \frac{e^{\tilde{s}_i}}{\sum_{j=1}^{N+1} e^{\tilde{s}_j}}. \tag{10}$$

Assuming that the training has been conducted M times, the entire training process involves samples from N different classes. For the backward propagation of any single training result, the result is as follows:

$$\frac{\partial \mathrm{Loss}}{\partial \hat{F}} = \begin{bmatrix} \frac{\partial \mathrm{Loss}}{\partial \tilde{f}_1} \\ \vdots \\ \frac{\partial \mathrm{Loss}}{\partial \tilde{f}_N} \\ \frac{\partial \mathrm{Loss}}{\partial \tilde{f}_{N+1}} \end{bmatrix} = \begin{bmatrix} C_1^* A^{\mathrm{T}} \left(\frac{e^{\tilde{s}_1}}{\sum_{j=1}^{N+1} e^{\tilde{s}_j}} - y_1 \right) \\ \vdots \\ C_N^* A^{\mathrm{T}} \left(\frac{e^{\tilde{s}_N}}{\sum_{j=1}^{N+1} e^{\tilde{s}_j}} - y_N \right) \\ \frac{1}{N} \sum_{i=1}^{N} C_i^* A^{\mathrm{T}} \left(\frac{e^{\tilde{s}_{N+1}}}{\sum_{j=1}^{N+1} e^{\tilde{s}_j}} - y_{N+1} \right) \end{bmatrix}. \tag{11}$$

Expanding the gradient update process for classifiers $\tilde{f}_j$ of different classes, where t is the number of iterations, the entire iteration process is as follows:

$$\begin{aligned} \tilde{f}_1 &= f_c + \eta C_1^\star A_1^T \tilde{\delta}_{1,1} + \cdots + \eta C_1^\star A_N^T \tilde{\delta}_{1,M} \\ &\cdots \\ \tilde{f}_N &= f_c + \eta C_N^\star A_1^T \tilde{\delta}_{N,1} + \cdots + \eta C_N^\star A_N^T \tilde{\delta}_{N,M} \\ \tilde{f}_{N+1} &= f_c + \eta C_{N+1}^\star \left[A_1^T \tilde{\delta}_{N+1,1} + \cdots + A_N^T \tilde{\delta}_{N+1,M} \right] \end{aligned} \tag{12}$$

where $f_c \sim N(0,1)$ represents the weight parameters initialized by Kaiming initialization, η represents the learning rate. $\widetilde{\delta}_{j,t}$ indicates the discrepancy between the predicted class and the true value. It is assumed that $A_i^T \widetilde{\delta}_{j,t} \in \mathbb{R}^{d\times 1}$ and C_j follows the Gaussian distribution $A_i^T \widetilde{\delta}_{j,t} \sim N(\tilde{\mu}_{j,i,t}, \tilde{\sigma}_{j,i,t})$, $C_j \sim N(\mu_{C_j}, \sigma_{C_j})$. According to the calculation rules of the Gaussian distribution [44], we can obtain

$$\begin{aligned} \mu_{\widetilde{f}_j} &= \sum_{i\in N, t=1}^{M} \frac{\widetilde{\mu}_{j,i,t}\sigma_{C_j}^2 + \mu_{C_j}\widetilde{\sigma}_{j,i,t}^2}{\sigma_{C_j}^2 + \widetilde{\sigma}_{j,i,t}^2} \\ \sigma_{\widetilde{f}_j} &= 1 + \sum_{i\in N, t=1}^{M} \frac{\eta^2}{1/\sigma_{C_j}^2 + 1/\widetilde{\sigma}_{j,i,t}^2}. \end{aligned} \tag{13}$$

In few-shot scenarios, there is a limited number of samples per class, while background samples are not scarce. Therefore, it is reasonable to assume that the classification network can identify background features relatively accurately, resulting in a large number of 0s in $\widetilde{\delta}_{j,t}$, which leads to $A_i^T \widetilde{\delta}_{j,t} \sim N(\tilde{\mu}_{j,i,t}, \tilde{\sigma}_{j,i,t})$ having characteristics of low mean and low variance. Based on the above analysis, we calculate the differences between the distributions of any two classifiers

$$\begin{aligned} &\mu_{\widetilde{f}_j} - \mu_{\widetilde{f}_p} \\ &= \sum_{i\in N, t=1}^{M} \frac{\widetilde{\mu}_{j,i,t}\sigma_{C_j}^2 + \mu_{C_j}\widetilde{\sigma}_{j,i,t}^2}{\sigma_{C_j}^2 + \widetilde{\sigma}_{j,i,t}^2} - \frac{\widetilde{\mu}_{p,i,t}\sigma_{C_p}^2 + \mu_{C_p}\widetilde{\sigma}_{p,i,t}^2}{\sigma_{C_p}^2 + \widetilde{\sigma}_{p,i,t}^2} \\ &\approx \sum_{i\in N, t=1}^{M} \frac{\mu_{C_j}\widetilde{\sigma}_{j,i,t}^2}{\sigma_{C_j}^2} - \frac{\mu_{C_p}\widetilde{\sigma}_{p,i,t}^2}{\sigma_{C_p}^2}, \quad \widetilde{\mu}_{*,i,t} \to 0 \& \sigma_{C_*}^2 \gg \widetilde{\sigma}_{*,i,t}^2 \end{aligned}$$

$$\begin{aligned} &\sigma_{f_j} - \sigma_{f_p} \\ &= \sum_{i\in N, t=1}^{M} \frac{\eta^2}{1/\sigma_{C_j}^2 + 1/\sigma_{j,i,t}^2} - \frac{\eta^2}{1/\sigma_{C_p}^2 + 1/\sigma_{p,i,t}^2} \\ &\approx \sum_{i\in N, t=1}^{M} \eta^2 \left(\widetilde{\sigma}_{j,i,t}^2 - \widetilde{\sigma}_{p,i,t}^2 \right), \widetilde{\mu}_{*,i,t} \to 0 \& \sigma_{C_*}^2 \gg \widetilde{\sigma}_{*,i,t}^2. \end{aligned} \tag{14}$$

Similarly, the difference between different classifiers in Meta R-CNN can be obtained

$$\begin{aligned} &\mu_{f_j} - \mu_{f_p} \\ &= \sum_{i\in N, t=1}^{M} \frac{\mu_{j,i,t}\sigma_{C_i}^2 + \mu_{C_i}\sigma_{j,i,t}^2}{\sigma_{C_i}^2 + \sigma_{j,i,t}^2} - \frac{\mu_{p,i,t}\sigma_{C_i}^2 + \mu_{C_i}\sigma_{p,i,t}^2}{\sigma_{C_i}^2 + \sigma_{p,i,t}^2} \\ &\approx \sum_{i\in N, t=1}^{M} \frac{\mu_{C_i}}{\sigma_{C_i}^2} \left(\sigma_{j,i,t}^2 - \sigma_{p,i,t}^2 \right), \quad \mu_{*,i,t} \to 0 \& \sigma_{C_*}^2 \gg \sigma_{*,i,t}^2 \\ &\sigma_{f_j} - \sigma_{f_p} \\ &= \sum_{i\in N, t=1}^{M} \frac{\eta^2}{1/\sigma_{C_i}^2 + 1/\sigma_{j,i,t}^2} - \frac{\eta^2}{1/\sigma_{C_i}^2 + 1/\sigma_{p,i,t}^2} \\ &\approx \sum_{i\in N, t=1}^{M} \eta^2 \left(\sigma_{j,i,t}^2 - \sigma_{p,i,t}^2 \right), \quad \mu_{*,i,t} \to 0 \& \sigma_{C_*}^2 \gg \sigma_{*,i,t}^2 \end{aligned} \tag{15}$$

where i represents the class of the input sample, and j (or p) represents the class of the classifier. The differences in classifier distributions in Meta R-CNN are due to two factors: the feature prototypes of input sample classes $(\mu_{C_i})/(\sigma_{C_i}^2)$ and the classification errors of the classifiers on current samples $\sigma_{j,i,t}^2, \sigma_{p,i,t}^2$. In Meta R-CNN, since classifier distribution differences are related to input sample classes, classifiers are more affected by high-frequency samples. This reduces the differences between classifiers and limits the model's adaptability and robustness. In contrast, UFEM not only focuses on classification errors but also incorporates feature prototypes corresponding to classifiers $(\mu_{C_j})/(\sigma_{C_j}^2), (\mu_{C_p})/(\sigma_{C_p}^2)$. As shown in Fig. 4(a), Meta R-CNN's classifier distribution differences are prone to high-frequency sample interference, leading to concentrated distributions and poor adaptation to low-frequency or novel classes. As shown in Fig. 4(b), by integrating classifier-specific feature prototypes, UFEM effectively reduces the impact of individual input samples. This maintains the diversity of classifier distributions, thereby enhancing the model's classification performance and generalization ability in complex scenarios. Consequently, UFEM demonstrates greater robustness and adaptability when processing diverse samples.

E. Gradient-Controlled Anti-Forgetting Optimizer

In few-shot learning, catastrophic forgetting refers to the decline in detection and recognition performance of base classes after novel classes are introduced during the subsequent training stage. The essence is that the updated gradients during the learning of novel classes affect the detection and recognition of base classes. If the impact of gradient updates on base classes during the learning of novel classes can be reduced, the problem of catastrophic forgetting can be alleviated.

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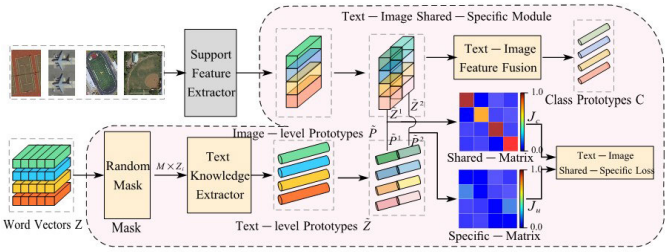


Fig. 3. Architecture of text–image SSM.

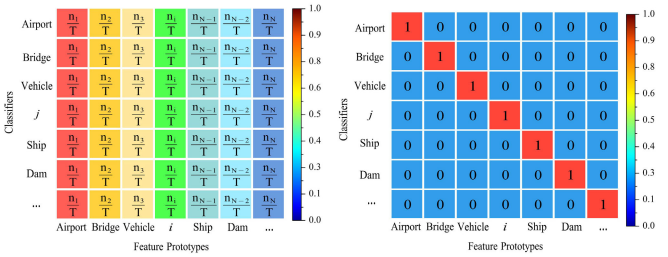


Fig. 4. Impact of feature prototypes. In (a), the numbers $(n_*)/T$ represent the frequency of samples in that class appearing during the training process. To better illustrate the impact of feature prototypes on the classifier in Meta R-CNN, it is assumed that the training samples decrease from left to right ("airport" has the most samples). (a) Meta R-CNN. (b) Ours.

During the pretraining stage, the base classes are sufficiently trained, so their gradient values will be relatively small. In the subsequent training stage, the gradients of the base classes can be used as an observer to adjust the gradients of the entire network.

In the subsequent training stage, decoupling is performed for base and novel classes to obtain the classifiers and regressors for base classes and novel classes. The weights for the classifiers and regressors of the base classes are initialized with the weights from the pretraining stage, while those for the novel classes are initialized with Kaiming initialization. It should be noted that during the subsequent training stage, the backbone is frozen, so the gradients for this part will be zero during gradient updates.

As shown in Fig. 5, calculating the gradients of each module to obtain the gradient Grad for all modules in the network during the subsequent training stage. Then, the statistical values of the gradients for the base classes $\mathring{G}_b$ are calculated as follows:

$$\mathring{G}_b = \left(\left\|G_b^{\text{cls}}\right\|_2 + \left\|G_b^{\text{reg}}\right\|_2\right) * 0.5 \tag{16}$$

where G_b^{cls} and G_b^{reg} represent the gradients of the base class classifiers and regressors, respectively, and $\|*\|_2$ denotes the two-norm.

After obtaining $\mathring{G}_b$, the gradients of the network's modules are adjusted using the following formula:

$$G_{\text{new}} = \text{Grad} \times e^{-\mathring{G}_b} \tag{17}$$

where G_{new} represents the updated gradients. If the current statistical value of the base class gradients $\mathring{G}_b$ is large, it indicates that if the parameters are updated according to the current gradients, the performance of the base classes is likely to decline. In this case, the GAO will reduce the gradients of the current network modules, alleviating the performance decline of the base classes.

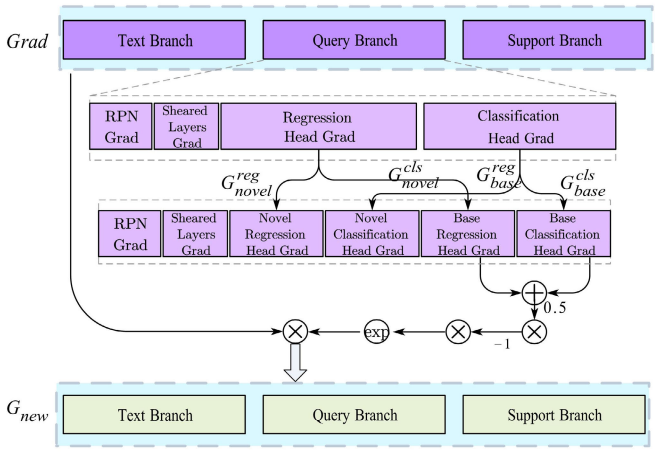


Fig. 5. Architecture of GAO.

IV. Experiment

A. Datasets

1) *DIOR:* The DIOR dataset is a large-scale optical remote sensing image object detection benchmark dataset, including 23 463 RSIs and 1 90 288 object instances. It covers 20 different object categories, such as airplanes, airports, baseball fields, basketball courts, etc., providing a rich set of training samples and a wide range of application scenarios. The spatial resolution varies from 0.5 to 30 m. Following the FSOD setup in [21] and [28], the dataset adopts four distinct novel/base class splits, each split containing 15 base classes and five novel classes. The novel classes for each partition are shown in Table I, and the remaining classes are considered base classes.
2) *NWPU VHR-10:* The NWPU VHR-10 dataset consists of 800 RSIs. The image sizes range from 500 × 500 to 1100 × 1100 pixels, with a total of 3651 target instances, and spatial resolutions vary from 0.5 to 30 m. The dataset covers ten different types of remote sensing targets, including airplanes, ships, storage tanks, baseball fields, tennis courts, basketball courts, ground track fields, ports, bridges, and vehicles. Similar to DIOR, the dataset adopts two distinct novel/base class splits, with each combination containing seven base classes and three novel classes. The novel classes for each partition are shown in Table I.
3) The text descriptions of the DIOR and NWPU VHR-10 datasets are derived from TEMO [13].

B. Evaluation Metrics

To validate the effectiveness of the method, we evaluate the detection results using mean average precision (mAP). mAP is a commonly used metric for evaluating the performance of object detection models, especially useful in multiclass

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TABLE I

DIFFERENT NOVEL CLASS SPLIT SETTINGS FOR FSIOD SCENARIOS

Dataset	Split	Novel Classes				
DIOR	1	Baseball field	Basketball court	Bridge	Chimney	Ship
	2	Airplane	Airport	Expressway toll station	Harbor	Ground track field
	3	Dam	Golf course	Storage tank	Tennis court	Vehicle
	4	Express service area	Overpass	Stadium	Train station	Windmill
NWPU VHR-10	1	Airplane	Baseball diamond	Tennis court		
	2	Basketball court	Ground track field	Vehicle		

TABLE II

ABLATION STUDY FOR THE DESIGNED COMPONENTS

Components				pretrain	10-shot			20-shot		
Only	Frozen E*	UFEM	SSM GAO	Base	Base	Novel	All	Base	Novel	All
				68.27	65.30	17.28	53.30	65.77	18.20	53.88
✓				68.27	60.89	38.07	55.20	61.33	40.15	56.04
✓			✓	68.27	64.30	37.95	57.71	64.69	39.66	58.43
✓	✓			69.20	62.58	38.03	56.45	62.33	43.48	57.62
✓	✓	✓		72.66	66.55	38.48	59.54	66.67	43.59	60.90
✓	✓	✓	✓	72.66	70.52	38.53	62.53	71.31	41.89	63.69

* E represents Feature Extractor.

problems. It assesses the overall precision of the model by calculating the average precision (AP) across all classes. AP is typically calculated by plotting the precision-recall curve and then calculating the area under the curve. A common approach is to use interpolation to compute the AP. mAP is the average of the AP values across all classes, representing the model's overall precision across all classes [21], [28].

C. Implementation Details

Our proposed UFEDet utilizes Meta R-CNN as the baseline. During the pretraining stage, we conducted 60 000 iterations with an initial learning rate of 0.02, which was adjusted to 0.002 at the 40 000 iteration and to 0.0002 at the 50 000 iteration. In the subsequent training stage, we froze the feature extractor and introduced GAO to adjust the gradients during backpropagation to prevent catastrophic forgetting of the base classes. All experiments were completed on a single NVIDIA 4090 GPU. During the testing stage, the confidence threshold and the IOU threshold for nonmaximum suppression were set to 0.05 and 0.5, respectively.

D. Ablation Study

To validate the effectiveness of our proposed UFEDet and various modules, we conducted ablation studies on the DIOR dataset.

1) Ablation Study for Each Component: We trained Meta R-CNN on the DIOR dataset as the baseline, with the mAP of the base classes during the pretraining stage being 68.27%. After subsequent training, the mAP for the novel classes reaches 17.28%, 18.20% in 10, 20-shot scenarios, respectively. To validate the effectiveness of the proposed UFEDet, we conducted ablation studies on each module, with results shown in Table II.

During the pretraining stage of Meta R-CNN, novel classes were treated as background. In the subsequent training stage, too many weights were frozen, which, although preventing catastrophic forgetting, led to poor performance on novel classes because the frozen modules did not receive gradient updates during training. Inspired by [13], we only froze the feature extractor in the subsequent training stage. As a result, the performance of few-shot classes improved, but there was

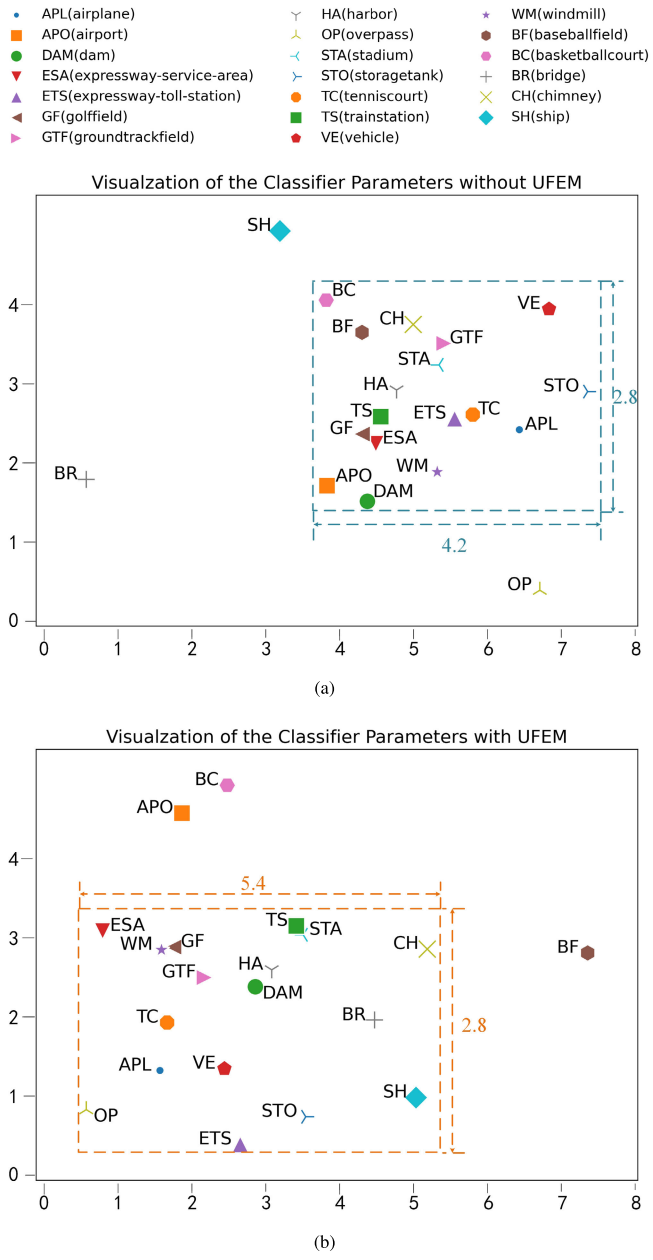


Fig. 6. Visualize the classifier parameters after dimensionality reduction using factor analysis. (a) Without UFEM, most classifiers are tightly clustered within a 4.2×2.8 region (blue box). (b) With UFEM, the classifiers are more widely dispersed across a 5.4×2.8 region (orange box), demonstrating that UFEM effectively enhances parameter diversity and inter-class separability.

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TABLE III
ABLATION STUDY FOR CROSS-MODAL ASSEMBLY MODULE

stage	pretrain	20-shot		
	Base	Base	Novel	All
Meta RCNN w UFEM	69.20	62.33	43.48	57.62
structure 1	68.10	66.01	37.15	58.80
structure 2	70.20	63.50	39.48	57.49
structure 3	**72.66**	**66.67**	**43.59**	**60.90**

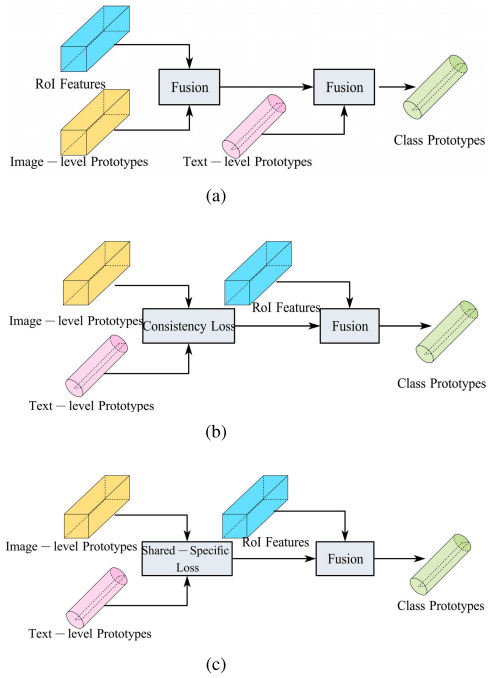


Fig. 7. Different assembly methods for fusing image-level prototypes and text-level prototypes to generate class prototypes. (a) Approach 1. (b) Approach 2. (c) Approach 3.

a corresponding performance loss for the base classes. The GAO module introduced during the subsequent training stage significantly reduced the performance loss of the base classes, demonstrating its effectiveness. To further enhance detection and recognition performance, particularly for few-shot classes, we introduced the UFEM module. On the one hand, UFEM improves class discriminability by leveraging class feature prototypes, and on the other hand, it incorporates textual description information to mitigate the information deficiency caused by the scarcity of image samples. The experimental results show that the introduction of both the UFEM and SSM modules not only improved detection accuracy for base classes but also enhanced detection and recognition precision for novel classes. Moreover, the two modules complement each other and can further enhance performance when used together. Overall, the experimental results validate the effectiveness of the proposed modules, which improve performance for both novel and base classes while significantly alleviating catastrophic forgetting.

2) Ablation Study for UFEM: To further demonstrate that the UFEM effectively amplifies the differences between classifier parameters, we visualized the classifier parameters in Fig. 6. Following the approach in [16], we applied factor analysis [36] for dimensionality reduction of the parameters, as factor analysis performs better when visualizing a small

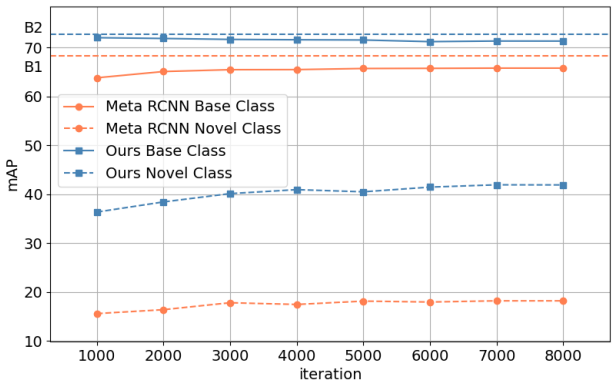


Fig. 8. Ablation study for GAO.

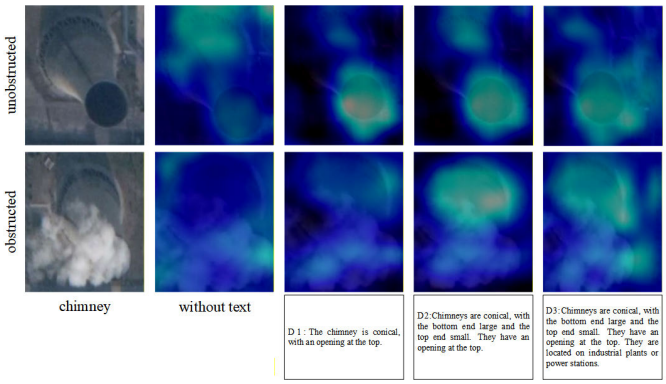


Fig. 9. Activation maps for each description.

amount of data [16]. As shown in the blue-boxed area in Fig. 6(a), without the UFEM, most classifiers are clustered in a 4.2 × 2.8 area. In contrast, as shown in the orange-boxed area in Fig. 6(b), after applying the UFEM, most classifiers are spread across a 5.4 × 2.8 area. Thus, from a visual perspective, the UFEM causes the classifiers to be more dispersed overall, indicating that UFEM increases the parameter differences between different classifications. This result is consistent with the theoretical findings we derived.

3) Ablation Study for SSM: In our research on supplementing image information with textual information, we explored three approaches, as shown in Fig. 7. The first approach, shown in Fig. 7(a), is based on [13] and constructs a multilevel enhanced ROI feature enhancement model using both image-level and text-level prototypes. The second approach, as shown in Fig. 7(b), imposes consistency constraints on the image-level and text-level prototypes before enhancing the ROI features. The third approach, shown in Fig. 7(c), is the SSM module. We conducted experiments on split 1 of the DIOR dataset in the 20-shot setting, and the results are presented in Table III. It is clear that the first approach did not improve the performance of novel classes. We believe this is due to the heterogeneity between features from different modalities, where directly fusing them with ROI features may cause conflicts. Based on the results of the first approach, we designed the second approach, introducing consistency loss constraints. The experimental results show that, although there is still some performance loss for novel classes, the decline is smaller compared to the first approach. We believe that features from different modalities cannot be simply constrained by consistency loss, as each modality contains both

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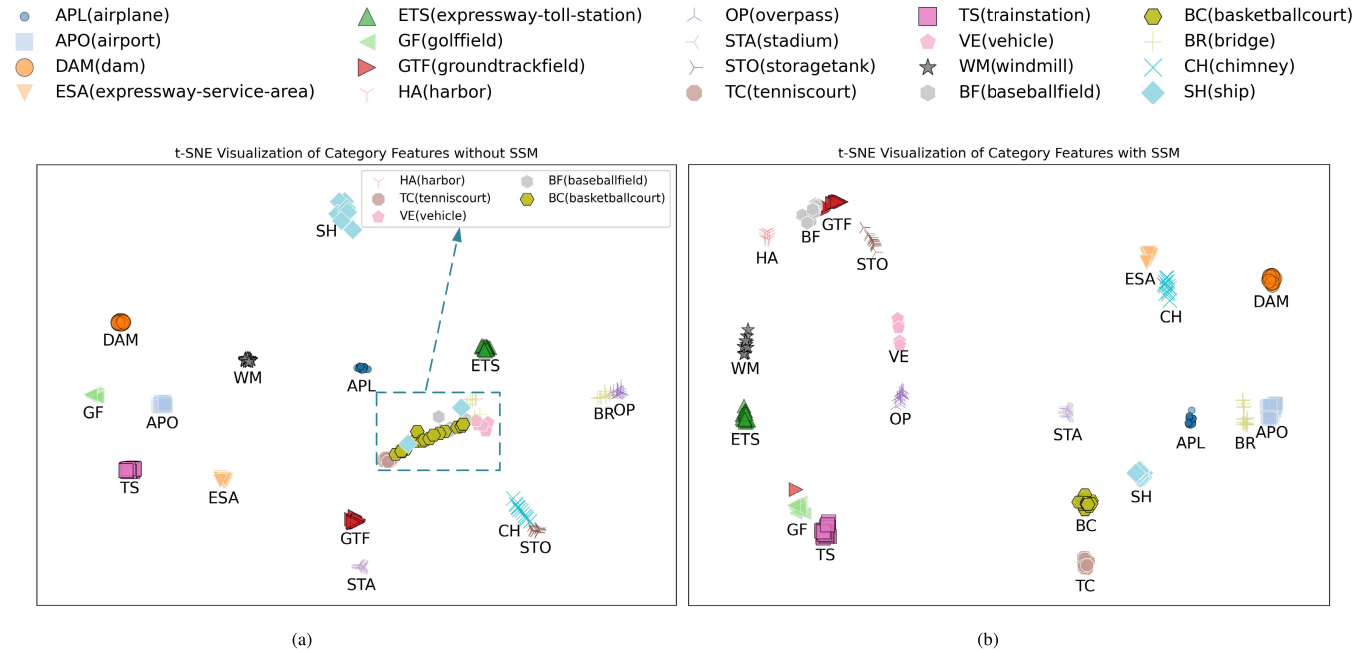


Fig. 10. Visualization of the class features in the 20-shot setting on the split 1 of DIOR. (a) Meta R-CNN. (b) Ours.

TABLE IV
DIFFERENT TEXTUAL DESCRIPTIONS OF AN AIRPLANE

Description	Textual Descriptions	Information Increment
1	An airplane typically has symmetrical wings distributed on both sides of the fuselage.	Shape-only descriptions
2	An airplane includes fuselage, wings, tail, and engines, etc. The two wings of the aircraft are symmetrically distributed on both sides of the fuselage.	Comprehensive attribute descriptions
3	An airplane includes fuselage, wings, tail, and engines, etc. The two wings of the aircraft are symmetrically distributed on both sides of the fuselage. It is typically adjacent to runways at airports, surrounded by taxiways and terminal buildings.	Attribute-enhanced spatial associations

TABLE V
ABLATION STUDY OF NUMBER OF DESCRIPTIONS

Description	3-shot	5-shot	10-shot	20-shot	Average Number of Sentences
Without Text	31.4	33.7	38.0	40.9	0
1	31.7	33.9	38.1	40.9	2.0
2	32.8	35.2	38.5	41.8	3.67
3	32.4	33.9	38.4	41.5	5.01

shared and unique features. Therefore, we proposed the SSM module, which extracts common features between modalities while retaining the unique features of each modality. The final experimental results of the SSM module show that the performance of novel classes has improved.

4) Ablation Study for Gao: Fig. 8 illustrates the comparative mAP dynamics for base and novel classes during subsequent training for Meta R-CNN (orange) and our GAO (blue). The straight dashed orange (B1) and blue lines (B2) represent the pretraining mAP for base classes, while the solid curves indicate the mAP during subsequent training for the base classes. The dashed curves represent the mAP for novel classes. Our method shows a smaller mAP decline for base classes and higher mAP for novel classes compared to Meta R-CNN throughout subsequent training.

TABLE VI
ABLATION STUDY OF TEXT–IMAGE ALIGNMENT

Setting	3-shot	5-shot	10-shot	20-shot
w/ Text Branch (D2)	32.8	35.2	38.5	41.8
w/o Text Branch	31.4	33.7	38.0	40.9
Random Text	30.8	31.7	36.9	39.6

5) Ablation Study of the Number of Descriptions: To investigate the impact of the number of descriptions on FSOD performance, we propose three progressively detailed description strategies: shape-only descriptions, descriptions incorporating component spatial layouts, and descriptions further enriched with relationships to surrounding objects—taking "airplane" as an example, as shown in Table IV. The average number of textual descriptions per image in these three strategies is 2.0, 3.67, and 5.01, respectively.

We evaluated these descriptions under 3-shot, 5-shot, 10-shot, and 20-shot settings. As shown in Table V, detection performance gradually improves as the descriptions evolve from coarse to more structurally informative: shape-focused cues in Description 1 provide a clear benefit over the no-text baseline, while the component- and layout-aware structural descriptions in Description 2 bring further consistent gains across different shot settings. However, extending the descriptions to include richer contextual relationships in Description 3 leads to only marginal, and in some cases unstable, improvements.

To better understand these behaviors, we visualized feature maps under different textual descriptions. As shown in Fig. 9, the first row corresponds to the chimney without smoke obstruction, and the second row to the same scene under

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TABLE VII
UFEDET PERFORMANCE ON THE DIOR DATASET

Split	Method	Year	3-shot			5-shot			10-shot			20-shot		
			Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	FRW	2021	49.40	7.50	38.90	49.70	12.10	40.30	49.50	18.10	41.70	50.00	22.00	43.00
	Meta RCNN	2019	60.62	12.02	48.47	62.01	13.09	49.78	65.30	17.28	53.30	65.77	18.20	53.88
	FsDetView	2023	59.54	13.19	47.95	58.58	14.29	47.51	59.64	18.02	49.24	62.69	18.01	51.52
	P-CNN	2022	47.00	18.00	39.80	48.40	22.80	42.00	50.90	27.60	45.10	52.20	29.60	46.80
	TFA w/cos	2020	**70.32**	11.35	55.58	70.51	11.57	55.78	70.52	15.37	56.73	71.07	17.96	57.79
	TEMO	2023	60.89	28.04	52.68	64.30	31.05	55.99	65.53	35.38	57.99	65.62	40.25	59.28
	iMTFA	2021	60.13	23.34	50.93	62.51	25.20	53.18	65.81	29.35	56.70	62.43	35.22	55.63
	Incremental-DETR	2023	65.93	27.16	56.24	66.19	30.21	57.20	69.98	36.44	61.60	70.13	37.55	61.99
	G-FSDet	2023	68.94	27.57	58.61	69.52	30.52	59.72	69.03	37.46	61.16	69.80	39.83	62.31
	AMTN	2024	45.5	22.0	39.63	47.6	24.7	41.2	49.7	29.5	44.65	52.0	32.2	47.05
	CCFR	2024	50.5	21.9	43.35	55.2	35.4	50.25	54.7	40.7	51.2	59.4	42.0	55.05
	SMDC	2025	69.7	18.4	56.9	70.0	21.8	58.0	70.9	25.4	59.5	71.0	30.0	60.7
	MFKDIOD	2025	65.7	33.2	57.58	67.1	**37.7**	59.75	68.8	**41.7**	62.02	69.9	**43.5**	63.3
	TPG-FSOD	2025	62.7	**34.2**	55.6	62.7	35.6	55.9	64.4	40.2	58.4	65.5	43.0	59.9
	Ours		70.23	32.89	**60.9**	**70.64**	35.20	**61.8**	**70.52**	38.53	**62.5**	**71.31**	41.89	**64.0**
2	FRW	2021	48.50	4.80	37.60	46.80	7.00	36.90	46.40	9.00	37.10	43.50	14.10	36.20
	Meta RCNN	2019	62.55	8.84	49.12	63.14	10.88	50.07	63.28	14.90	51.18	63.86	16.71	52.07
	FsDetView	2023	58.88	10.83	46.87	60.31	9.63	47.64	61.16	13.57	49.26	61.16	14.76	49.56
	P-CNN	2022	48.90	14.50	40.30	49.10	14.90	40.60	52.50	18.90	44.10	51.60	22.80	44.40
	TFA w/cos	2020	70.75	5.77	54.51	70.79	8.19	55.14	69.93	8.71	54.63	70.02	12.18	55.56
	TEMO	2023	61.39	15.57	49.94	65.23	17.98	53.42	62.62	21.04	52.23	61.18	23.04	51.65
	iMTFA	2021	59.97	13.13	48.26	60.26	14.92	48.93	60.35	19.37	50.11	61.32	23.32	51.82
	Incremental-DETR	2023	67.29	12.57	53.61	66.98	15.04	54.00	67.15	20.10	55.39	68.03	20.17	56.07
	G-FSDet	2023	69.20	14.13	55.43	69.25	15.84	55.87	68.71	20.70	56.70	68.18	22.69	56.86
	AMTN	2024	48.1	15.1	39.85	50.0	18.1	42.02	51.1	20.6	43.48	52.3	24.7	45.40
	CCFR	2024	51.9	10.4	42.53	56.8	12.8	45.8	56.0	18.3	45.58	59.3	28.5	51.6
	MFKDIOD	2025	66.0	**20.0**	54.50	66.5	**21.8**	55.32	68.0	**28.8**	**58.20**	69.6	**29.5**	**59.58**
	TPG-FSOD	2025	61.3	14.3	49.6	61.8	19.0	51.1	63.0	24.8	53.5	63.3	25.6	53.9
	Ours		**71.48**	16.91	**57.39**	**71.30**	19.37	**57.77**	**70.95**	20.27	57.85	**71.60**	23.00	59.04
3	FRW	2021	45.50	7.80	36.10	47.90	13.70	39.30	44.50	13.80	36.80	43.50	18.50	37.30
	Meta RCNN	2019	61.93	9.10	48.72	63.44	12.29	50.66	62.57	11.96	49.92	65.53	16.14	53.1
	FsDetView	2023	61.00	7.49	47.63	61.33	12.61	49.15	61.94	11.49	49.32	65.17	17.02	53.14
	P-CNN	2022	49.50	16.50	41.30	49.90	18.80	42.10	52.10	23.30	44.90	53.10	28.80	47.0
	TFA w/cos	2020	71.95	8.36	56.05	71.64	10.13	56.26	72.56	10.75	57.11	**73.13**	17.99	59.3
	TEMO	2023	65.88	16.02	53.42	65.76	22.83	55.03	67.75	24.68	56.98	67.74	30.57	58.45
	iMTFA	2021	63.78	15.93	51.82	64.33	20.15	53.29	65.55	23.46	55.03	65.93	27.85	56.4
	Incremental-DETR	2023	68.33	16.35	55.34	68.84	22.78	57.33	69.35	25.03	58.27	67.83	31.34	58.78
	G-FSDet	2023	71.10	16.03	57.34	70.18	23.25	58.43	71.08	26.24	59.87	71.26	**32.05**	61.46
	AMTN	2024	47.1	18.5	39.95	48.5	20.5	41.5	51.7	23.7	44.70	63.5	29.2	54.93
	CCFR	2024	61.2	19.8	50.85	56.4	22.9	48.02	60.4	25.3	51.63	64.2	35.2	56.95
	MFKDIOD	2025	67.5	21.6	56.02	68.0	24.9	57.23	69.6	27.6	59.10	70.3	31.5	60.60
	TPG-FSOD	2025	65.6	20.1	54.2	65.1	23.1	54.6	66.4	28.9	57	65.9	32.6	57.6
	Ours		**73.16**	**22.74**	**60.60**	**73.05**	**26.12**	**61.32**	**72.95**	**29.48**	**62.13**	72.64	31.87	**62.40**
4	FRW	2021	48.20	3.70	37.10	48.50	6.80	38.10	45.70	7.20	36.10	44.40	12.20	36.40
	Meta RCNN	2019	61.73	13.94	49.78	62.60	15.84	50.91	62.23	15.07	50.44	63.24	18.17	51.98
	FsDetView	2023	58.90	14.28	47.75	58.97	15.95	48.22	60.37	15.37	49.12	60.89	16.96	49.91
	P-CNN	2022	49.80	15.20	41.20	49.90	17.50	41.80	51.70	18.90	43.50	52.30	25.70	45.70
	TFA w/cos	2020	68.57	10.42	54.03	68.85	14.29	55.21	68.58	14.35	55.03	68.86	12.01	54.65
	TEMO	2023	66.35	17.13	54.05	66.30	**21.28**	55.05	66.82	26.77	56.81	67.24	32.08	58.45
	iMTFA	2021	63.39	15.44	51.40	60.49	19.75	50.31	64.97	24.32	54.81	62.03	31.00	54.27
	Incremental-DETR	2023	67.05	14.48	53.91	66.31	20.55	54.87	64.38	25.65	54.70	66.37	28.49	56.90
	G-FSDet	2023	**69.01**	16.74	55.95	67.96	21.03	56.30	68.55	25.84	57.87	67.73	31.78	58.75
	CCFR	2024	51.3	11.4	41.33	54.5	20.4	45.98	53.4	27.9	47.03	58.3	33.8	52.18
	MFKDIOD	2025	64.3	16.7	52.4	67.1	21.0	55.58	68.1	25.8	57.53	69.1	30.6	59.48
	TPG-FSOD	2025	60.1	13.1	50.9	62.3	22.0	52.2	60.7	26.9	52.3	60.6	31.0	53.2
	Ours		68.75	**19.20**	**56.41**	**69.49**	21.07	**57.43**	**69.12**	**27.02**	**58.60**	**69.93**	**33.28**	**60.77**

smoke, each visualized under four conditions: without text, Description 1, Description 2, and Description 3. Without textual guidance, the model relies purely on visual cues, yielding only weak activations on the chimney in the clear case and almost none under smoke, revealing its limited ability to perceive occluded targets. With Description 1, which describes the upper opening and overall shape, the activation around the chimney top is noticeably enhanced in the unobstructed case, but quickly degrades when smoke obscures the opening, indicating that such partial shape cues offer only limited robustness to occlusion. In contrast, Description 2, which additionally encodes detailed upper and lower structural forms, leads to strong and more spatially coherent activations over both ends of the chimney without smoke; under smoke, although the top response weakens, the bottom region remains strongly activated, demonstrating improved resilience to partial occlusion through richer structural alignment between text and vision. When Description 3 is used, the activation pattern

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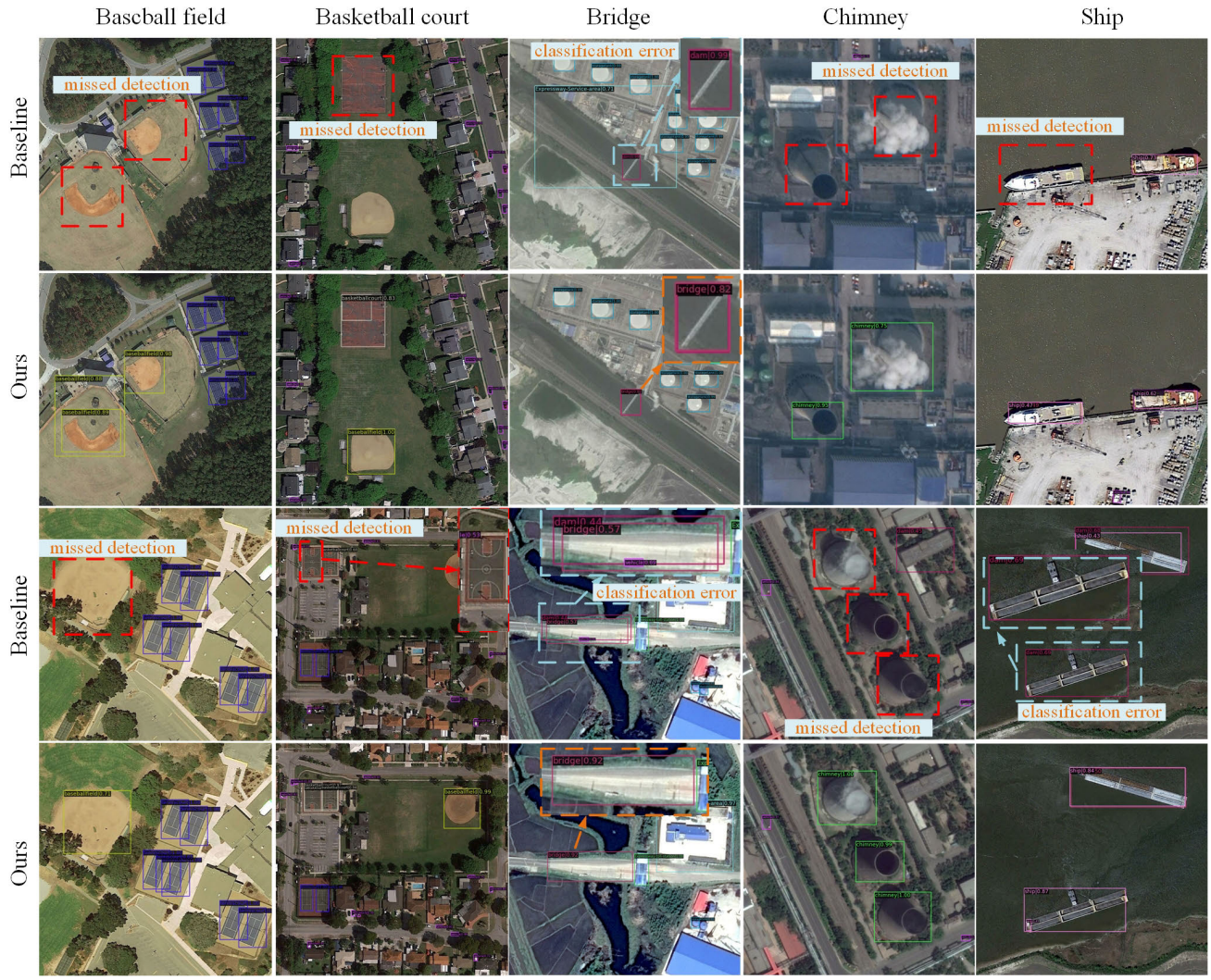


Fig. 11. Visualization comparison between baseline and UFEDet on DIOR dataset of 20-shot setting.

shifts: responses on the chimney itself become weaker, while those on the surrounding environment (e.g., nearby buildings or background regions) are amplified in both clear and occluded cases. This confirms that excessive, environment-focused contextual information diverts attention away from the target and dilutes the discriminative power of target-specific features.

These results confirm that well-targeted structural descriptions (particularly Description 2) significantly strengthen visual-text alignment and recognition robustness, whereas overly contextual descriptions (like Description 3) impair focus and accuracy.

6) Ablation Study of Text–Image Alignment: We systematically examined the effect of text–image alignment on FSOD performance through ablation studies.

As shown in Table VI, removing the text branch led to a significant drop in performance. Without the text branch, the model lost the mechanism for enriching visual features through textual cues, resulting in lower detection accuracy during inference. In addition, random text—such as "systems the bridge close and conical, large at the top, typically container are in surrounded school"—prevented the model from aligning the text with the image, further contributing to the decline in detection performance.

E. Comparisons With State-of-the-Art Methods

To demonstrate the effectiveness of our proposed UFEDet, we compared it with the state-of-the-art methods on the DIOR and NWPU VHR-10 datasets. The methods involved in the comparison include: FRW [22], Meta R-CNN [9], FsDetView [10], P-CNN [28], TFA [5], TEMO [13], iMTFA [38], Incremental-DETR [39], G-FSDet [40], AMTN [41], CCFR [33], SMDC [32], MFKDIOD [34], CAMCFormer [42], and TPG-FSOD [43].

1) Experiments on DIOR: Feature prototypes are crucial for few-shot learning. To validate the capability of our proposed method in extracting feature prototypes, we conducted experiments on split 1 of the DIOR dataset. We extracted features from 20 classes, with 20 samples per class. We used t-SNE to reduce the dimensionality of the feature prototypes, where each color represents a different class, and labeled the class name in the area where the samples of each class were most

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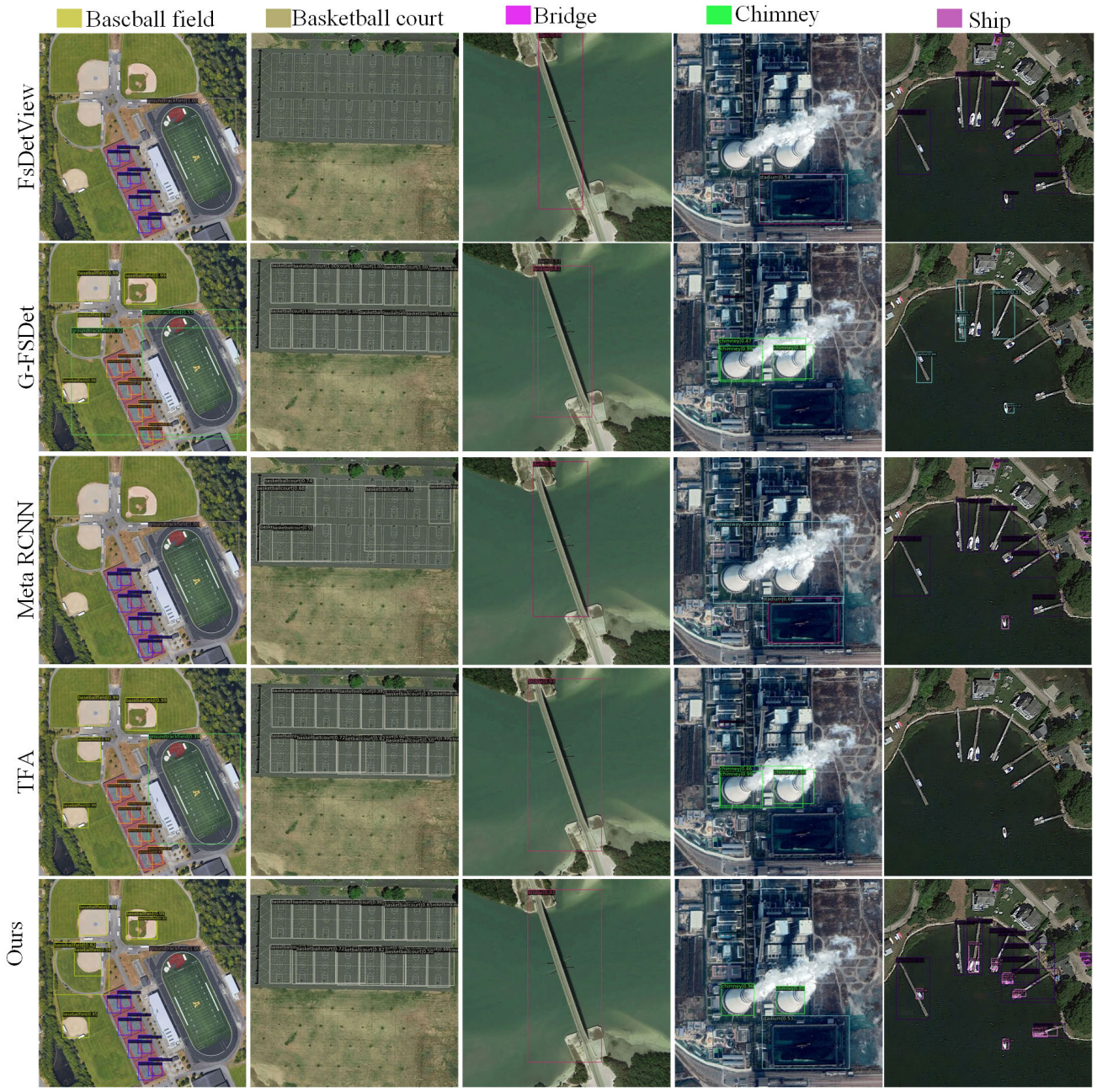


Fig. 12. Visualization comparison on DIOR dataset of 20-shot setting.

concentrated, as shown in Fig. 10. Ideally, samples from the same class should cluster together in the same area, and samples from different classes should be well-separated. As shown in Fig. 10(a), the feature prototypes extracted by the baseline method show mixing, as indicated in the blue boxed area of Fig. 10(a), where at least five classes are mixed together. This suggests that the model struggles to extract the unique features of each class, which negatively impacts detection performance. In contrast, as shown in Fig. 10(b), the feature prototypes for the 20 classes are clearly distinguishable, indicating that the model can successfully extract the unique features of each class, thus laying the foundation for subsequent ROI feature enhancement.

In Table VII, we compare our proposed method with state-of-the-art methods on the DIOR dataset across four few-shot scenarios: 3-shot, 5-shot, 10-shot, and 20-shot. An ideal FSOD method should not only improve performance on novel classes but also maintain strong performance on base classes. As shown in Table VII, our method achieves the best performance on novel classes in most scenarios. In addition, on base classes, our method performs excellently, with only slight performance drops compared to TFA and MFKDIOD in a few scenarios. This highlights the reliability of our proposed method.

Fig. 11 shows the results of baseline and our method on the novel classes of DIOR's split 1. It is evident that the baseline model suffers from both missed detections and classification

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TABLE VIII
UFEDET PERFORMANCE ON THE NWPU VHR-10 DATASET

Split	Method	Year	3-shot			5-shot			10-shot			20-shot		
			Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	FRW	2021	83.13	15.35	62.80	82.78	16.24	62.82	83.89	24.00	65.92	82.80	27.16	66.11
	Meta RCNN	2019	87.00	20.51	67.05	85.74	21.77	66.55	87.01	26.98	69.00	87.29	28.24	69.57
	FsDetView	2023	87.68	24.56	68.75	87.77	29.55	70.31	87.75	31.77	70.96	87.83	32.73	71.30
	P-CNN	2022	82.84	41.80	70.53	82.89	49.17	72.79	83.05	63.29	78.11	83.59	66.83	78.55
	TFA w/cos	2020	89.35	8.80	65.19	89.60	9.49	64.65	89.95	9.26	65.74	89.62	10.83	65.98
	DeRFCN	2021	87.22	37.90	73.33	86.87	46.08	74.63	87.29	62.95	79.99	87.26	64.61	80.46
	TEMO	2023	86.35	57.57	77.72	85.29	66.77	79.73	87.11	75.07	83.49	89.22	75.89	85.22
	iMTFA	2021	86.78	43.20	73.71	86.03	49.95	75.21	87.24	70.01	82.07	87.93	72.78	83.39
	Incremental-DETR	2023	87.16	50.93	76.29	87.05	53.18	76.89	87.89	69.94	82.51	88.35	73.18	83.80
	G-FSDet	2023	89.11	49.05	77.01	88.37	56.10	78.64	88.40	71.82	83.43	89.73	75.41	85.44
	MFKDIOD	2025	86.3	58.2	79.28	87.9	69.5	83.30	88.7	73.1	84.80	89.7	78.3	86.85
	TPG-FSOD	2025	89.4	56.1	79.41	89.8	64.7	82.3	89.2	75.6	85.1	88.9	75.9	85.0
	CAMCFormer	2025	-	54.2	-	-	60.0	-	-	77.2	-	-	80.1	-
	Ours		**91.44**	**66.76**	**84.04**	**90.75**	**73.03**	**85.44**	**90.18**	**81.96**	**87.72**	**90.14**	**79.15**	**86.85**
2	FRW	2021	85.34	28.61	68.32	85.75	31.45	69.45	85.49	32.53	69.60	86.58	33.93	70.79
	Meta RCNN	2019	86.86	21.41	67.23	87.38	35.34	71.77	87.56	37.14	72.43	87.26	39.47	72.92
	FsDetView	2023	88.11	39.01	73.38	89.34	40.31	74.63	89.34	45.09	76.07	89.31	46.28	76.40
	P-CNN	2022	81.03	39.32	68.52	81.18	46.10	70.70	80.93	55.90	73.41	81.21	58.37	75.50
	TFA w/cos	2020	**90.14**	11.14	66.44	**91.19**	12.46	67.57	**90.79**	11.35	66.96	90.37	11.56	66.73
	DeRFCN	2021	88.47	39.19	73.69	88.17	45.56	75.29	88.25	54.05	77.99	88.05	57.38	78.83
	TEMO	2023	88.25	50.82	77.02	88.31	59.31	79.61	89.07	64.52	81.71	87.93	73.35	83.56
	iMTFA	2021	87.36	46.29	75.04	88.05	54.34	77.94	88.03	62.56	80.39	88.32	66.37	81.74
	Incremental-DETR	2023	89.12	47.35	76.59	89.37	57.24	79.73	88.87	65.63	81.90	89.25	70.58	83.65
	G-FSDet	2023	89.99	50.09	78.02	90.52	58.75	80.99	89.23	**67.00**	**82.56**	**90.61**	75.86	**86.13**
	TPG-FSOD	2025	90.1	48.7	77.7	89.8	**63.5**	**81.9**	89.3	69.4	83.3	90.2	75.9	85.9
	Ours		89.12	**54.13**	**78.62**	89.36	56.63	79.50	88.85	65.10	81.73	89.03	**77.20**	85.48

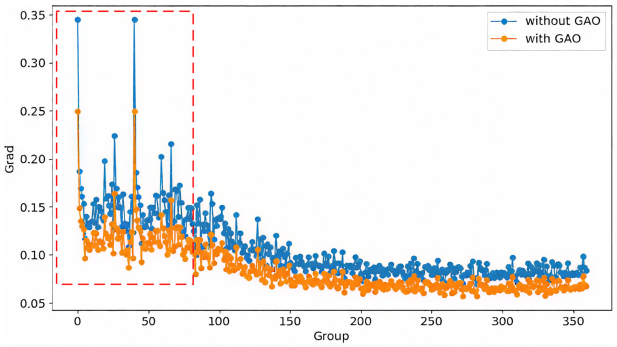


Fig. 13. Gradients without and with GAO.

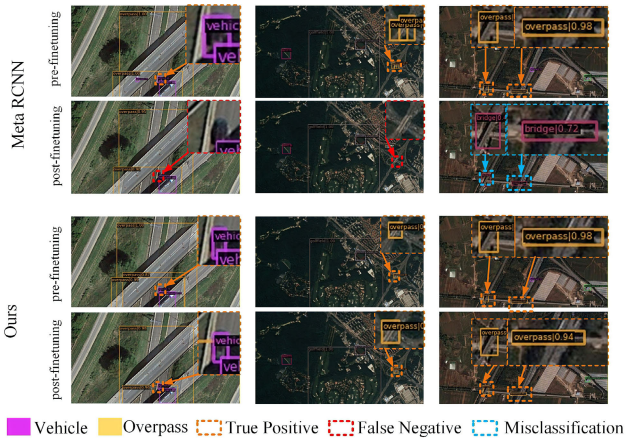


Fig. 14. Visualization of base classes before and after fine-tuning.

errors across all five novel classes. For instance, it fails to detect a basketball court, mistaking it for the background. In addition, it confuses a bridge with a dam because it does not effectively learn the distinguishing features of different classes. In contrast, our method mitigates these issues by enhancing the differences between classifier parameters and learning shared-specific features from both images and text, thereby reducing missed detections and classification errors.

Fig. 12 shows the results of other methods and our method on the novel classes of DIOR's split 1. As shown in Fig. 12, the comparison methods all exhibit missed detections on novel categories, especially when objects are occluded (Chimney) or small (Ship). This is due to the lack of image samples, causing the network to fail in adapting to complex scenarios. In contrast, our method incorporates textual information, achieving better detection performance even under partial occlusion and small object conditions (although our method also has limitations in these scenarios, which will be discussed in detail in Section IV-F1).

2) Experiments on NWPU VHR-10: To further verify the performance of our proposed method, we conducted experiments on the NWPU VHR-10 dataset, covering four few-shot scenarios: 3-shot, 5-shot, 10-shot, and 20-shot. The performance differences among various methods on the base classes of the NWPU VHR-10 dataset are not significant, which is due to the fact that the NWPU VHR-10 dataset has fewer images and lower scene complexity compared to the DIOR dataset. Under these circumstances, it is important to focus on the performance of novel classes. As shown in Table VIII, our method achieved the best performance on novel classes in most scenarios.

F. Discussion

1) Discussion on the Gradient Changes of the Base Classifiers: We visualize the gradients of base class classifiers

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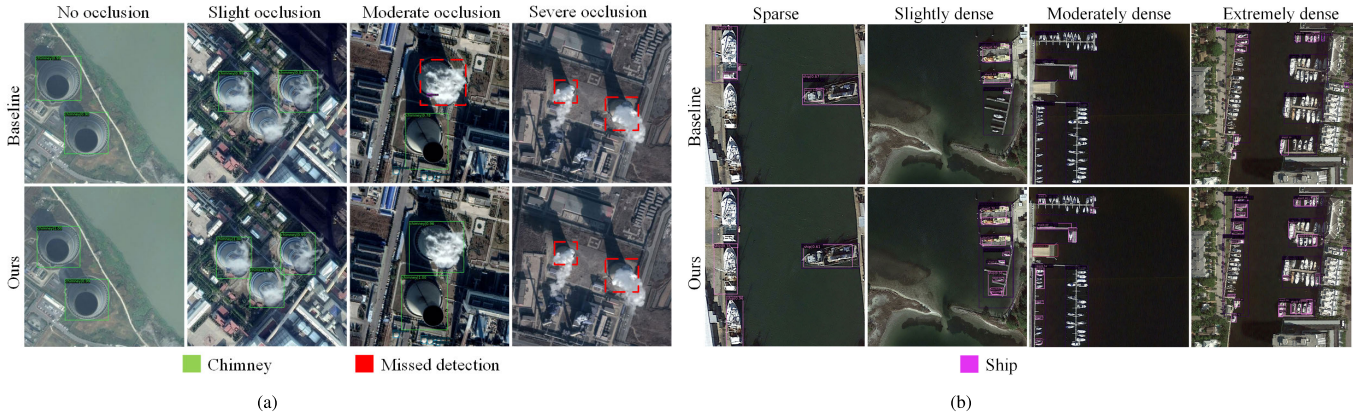


Fig. 15. Visualization of detection results in occluded and dense scenes. (a) Occluded scenes. (b) Dense scenes.

during training, with and without GAO, in Fig. 13. The horizontal axis shows iteration groups (50 iterations each) in the fine-tuning phase, and the vertical axis shows normalized gradient magnitudes. The blue and orange lines represent results without and with GAO, respectively.

As shown by the blue curve in the red dashed box in Fig. 13, the gradients of the base classes classifier without GAO exhibit significant changes during the fine-tuning stage. This occurs because novel class classifiers are randomly initialized, causing high initial prediction errors and strong gradient responses that disrupt the well-learned base class representations, leading to performance degradation [48], [49], [50].

To address this, we introduce the GAO module, which dynamically regulates gradient flows to reduce interference from novel class learning on base class features. As shown by the orange curve in Fig. 13, GAO significantly reduces gradient fluctuations for base classes, leading to more stable training and better preservation of base class performance.

Furthermore, we compare our method with the baseline in terms of base class detection performance before and after fine-tuning on few-shot classes. As shown in Fig. 14, the upper half of each row presents results before fine-tuning, and the lower half shows results after. After fine-tuning, the baseline suffers significant performance degradation, including missed detections of "vehicle" and "overpass," as well as misclassification of "overpass" as the novel class "bridge." In contrast, our method maintains strong performance on base classes, effectively mitigating catastrophic forgetting.

2) Analysis of Failure Cases: Current detection methods achieve high accuracy for base and novel classes, but still face challenges in complex occluded scenarios. For example, as shown in Fig. 15(a), our method successfully detects targets in scenarios without occlusion, slight occlusion, and moderate occlusion. The baseline performs well under no occlusion, but begins to miss detections under slight and moderate occlusion, demonstrating the effectiveness of our approach in complex scenarios. However, both methods suffer from missed detections under severe occlusion.

In densely populated and small target scenarios, few-shot detection also faces performance challenges. Fig. 15(b) illustrates this with a "ship" in harbors or crowded waters, where targets are severely occluded and occupy minimal pixel areas (e.g., ship). Our method detects ships effectively in sparse and slightly dense scenes, while the baseline works only in sparse scenes. In moderately and extremely dense scenes, although both methods experience missed detections, our method detects significantly more ships.

These problems stem from two main factors: 1) insufficient coverage of complex scenes in training data, which limits generalization ability, and 2) limitations in feature robustness and association modeling. Current methods lack effective feature completion for occluded regions and adequate optimization for multitarget association modeling (e.g., instance segmentation and contextual reasoning), making stable feature extraction in complex backgrounds challenging.

TABLE IX
COMPARISON ON THE COMPUTATION COST

	UFEDet	Meta RCNN
Inference time	0.063s	0.056s
Parameters	69.23M	60.73M
GFLOPs	1460.0	1390.0

3) Time and Space Complexity: We conducted a comparative analysis focusing on the computational overhead between our proposed UFEDet and Meta R-CNN, which is the most closely related approach. As detailed in Table IX, for an image resolution of $1333 \times 800 \times 3$ and utilizing a GTX 4090 GPU, the inference times recorded for UFEDet and Meta R-CNN were 0.063 and 0.056 s, respectively. Regarding the model parameters during inference, UFEDet and Meta R-CNN required 69.23 and 60.73 M, respectively. In addition, the computational complexity measured in GFLOPs was documented at 1460.0 for UFEDet and 1390.0 for Meta R-CNN. In summary, although UFEDet demands slightly more inference time and computational resources compared to Meta R-CNN, it delivers enhanced detection performance.

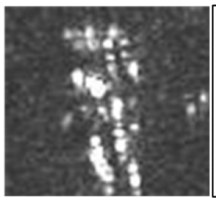
4) Limitations: The cross-modal transfer ability and applicability of our proposed method remain areas requiring further exploration. The most significant challenge lies in dataset limitations, particularly regarding the quality of textual descriptions. While the incorporation of textual descriptions

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There is a civilian airliner parked on the apron, appearing as a slender white object that contrasts sharply with the dark gray asphalt runway. The fuselage has a regular streamlined shape with distinctly swept wings.

(a)



The target exhibits a clear corner reflector effect, especially at the junctions of the nose, wings, and fuselage, where bright spots are formed. The overall contour is slender and slightly blurred in the azimuth direction.

(b)

Fig. 16. Differences in text descriptions between SAR and optical images. (a) Optical image and textual description. (b) SAR image and textual description.

has enhanced FSOD performance, their quality directly constrains the generalization of our approach. Crucially, visual characteristics of targets differ across modalities [45], [46], [47]: for instance, objects in SAR imagery are primarily characterized by scattering properties rather than optical features, as shown in Fig. 16.

V. Conclusion

FSOD for RSIs has received much attention. Three critical factors play a pivotal role in FSOD: features to be classified, reference features, and classifiers. However, existing methodologies have not comprehensively addressed these aspects. To fill this gap, we propose the UFEDet. Initially, we introduce the text–image SSM, which not only focuses on the consistent representation of the same class across different modalities but also emphasizes the unique information of each modality, thereby enhancing the distinctiveness of class feature prototypes. Subsequently, we propose an unconstrained feature enhancement module, which enhances the RoI features by leveraging the global class information from feature prototypes, while also increasing the distributional differences between classifiers. Finally, we introduce a gradient-controlled optimizer that modulates the gradient based on the gradient response of base classes, alleviating catastrophic forgetting of base classes during the training of novel classes. To validate the efficacy of our proposed approach, experiments were conducted on the DIOR and NWPU VHR-10 datasets. The experimental results demonstrate that the mAP of both base and novel classes exceeds that of popular few-shot object detectors.

References

[1] Z. Li et al., "Deep learning-based object detection techniques for remote sensing images: A survey," *Remote Sens.*, vol. 14, no. 10, p. 2385, May 2022.

[2] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[3] D. C. Schedl, I. Kurmi, and O. Bimber, "An autonomous drone for search and rescue in forests using airborne optical sectioning," *Sci. Robot.*, vol. 6, no. 55, p. 1188, Jun. 2021.

[4] S. Liu, Y. You, H. Su, G. Meng, W. Yang, and F. Liu, "Few-shot object detection in remote sensing image interpretation: Opportunities and challenges," *Remote Sens.*, vol. 14, no. 18, p. 4435, Sep. 2022.

[5] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9919–9928.

[6] J. Wu, S. Liu, D. Huang, and Y. Wang, "Multi-scale positive sample refinement for few-shot object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Glasgow, U.K.: Springer, Oct. 2020, pp. 456–472.

[7] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[8] B. Li, B. Yang, C. Liu, F. Liu, R. Ji, and Q. Ye, "Beyond max-margin: Class margin equilibrium for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7359–7368.

[9] G. Han, S. Huang, J. Ma, Y. He, and S.-F. Chang, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9577–9586.

[10] Y. Xiao, V. Lepetit, and R. Marlet, "Few-shot object detection and viewpoint estimation for objects in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3090–3106, Mar. 2023.

[11] H. Hu, S. Bai, A. Li, J. Cui, and L. Wang, "Dense relation distillation with context-aware aggregation for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10185–10194.

[12] G. Han, S. Huang, J. Ma, Y. He, and S.-F. Chang, "Meta Faster R-CNN: Towards accurate few-shot object detection with attentive feature alignment," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 780–789.

[13] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3250448.

[14] D. Bolya, S. Foley, J. Hays, and J. Hoffman, "TIDE: A general toolbox for identifying object detection errors," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2020, pp. 558–573.

[15] A. Papyan, R. M. M. Torki, and Y. N. Dauphin, "Feature learning in deep classifiers through intermediate neural collapse," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2202–2213, Jun. 2021.

[16] C. Chen, X. Yang, J. Zhang, B. Dong, and C. Xu, "Category knowledge-guided parameter calibration for few-shot object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 1092–1107, 2023.

[17] J. Han, Y. Ren, J. Ding, K. Yan, and G.-S. Xia, "Few-shot object detection via variational feature aggregation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 1, Jun. 2023, pp. 755–763.

[18] M. Guo, Y. You, and F. Liu, "Discriminative prototype learning for few-shot object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6425–6435, Aug. 2021.

[19] Y. Wang, C. M. Albrecht, N. A. A. Braham, C. Liu, Z. Xiong, and X. X. Zhu, "Decoupling common and unique representations for multimodal self-supervised learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 7, pp. 1889–1898, Jul. 2022.

[20] Z. Zhao, P. Tang, L. Zhao, and Z. Zhang, "Few-shot object detection of remote sensing images via two-stage fine-tuning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[21] Y. Wang, C. Xu, C. Liu, and Z. Li, "Context information refinement for few-shot object detection in remote sensing images," *Remote Sens.*, vol. 14, no. 14, p. 3255, Jul. 2022.

[22] X. Huang, B. He, M. Tong, D. Wang, and C. He, "Few-shot object detection on remote sensing images via shared attention module and balanced fine-tuning strategy," *Remote Sens.*, vol. 13, no. 19, p. 3816, Sep. 2021.

[23] R. Li, Y. Zeng, J. Wu, Y. Wang, and X. Zhang, "Few-shot object detection of remote sensing image via calibration," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2024, Art. no. 5603514.

[25] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601614.

[26] Y. Zhang, B. Zhang, and B. Wang, "Few-shot object detection with self-adaptive global similarity and two-way foreground stimulator in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7263–7276, 2022.

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[27] Z. Xiao, J. Qi, W. Xue, and P. Zhong, "Few-shot object detection with self-adaptive attention network for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4854–4865, 2021.
[28] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604610.
[29] J. Li, Y. Tian, Y. Xu, Z. Zhang, H. Wang, and Y. Xiao, "Meta-memory-based RCNN for few-shot object detection in remote sensing scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5635114.
[30] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603514.
[31] Y. Liu et al., "Few-shot object detection in remote-sensing images via label-consistent classifier and gradual regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612114.
[32] J. Wu, C. Qin, and G. Feng, "SMDC-Net: Saliency-guided multihead distribution calibration network for few-shot object detection on remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[33] W. Miao, Z. Zhao, J. Geng, and W. Jiang, "Few-shot object detection based on contrastive class-attention feature reweighting for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2800–2814, 2024.
[34] J. Chen, Y. Guo, D. Qin, J. Zhu, Z. Gou, and G. Sun, "Multiscale feature knowledge distillation and implicit object discovery for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5603215.
[35] Z. Zhou et al., "FSODS: A lightweight metalearning method for few-shot object detection on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5232217.
[36] M. Norris and L. Lecavalier, "Evaluating the use of exploratory factor analysis in developmental disability psychological research," *J. Autism Develop. Disorders*, vol. 40, no. 1, pp. 8–20, Jan. 2010.
[37] L. Qiao, Y. Zhao, Z. Li, X. Qiu, J. Wu, and C. Zhang, "DeFRCN: Decoupled faster R-CNN for few-shot object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8681–8690.
[38] D. A. Ganea, B. Boom, and R. Poppe, "Incremental few-shot instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1185–1194.
[39] N. Dong, Y. Zhang, M. Ding, and G. H. Lee, "Incremental-DETR: Incremental few-shot object detection via self-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 1, pp. 543–551.
[40] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.
[41] X. Chen et al., "Adaptive meta-knowledge transfer network for few-shot object detection in very high resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103675.
[42] L. Wang, S. Mei, Y. Wang, J. Lian, Z. Han, and Y. Feng, "CAMCFormer: Cross-attention and multicorrelation aided transformer for few-shot object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.
[43] T. Liu, S. Zhou, W. Li, Y. Zhang, and J. Guan, "Semantic prototyping with CLIP for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5615314.
[44] P. L. Green, "Distributed Gaussian processes with uncertain inputs," *IEEE Access*, vol. 12, pp. 176087–176093, 2024, Art. no. 176087.
[45] X. Shang et al., "CCLDet: A cross-modality and cross-domain low-light detector," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 3, pp. 3284–3294, Mar. 2025.
[46] L. Kong et al., "Few-shot class-incremental SAR target recognition via orthogonal distributed features," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 1, pp. 325–341, Feb. 2025.
[47] F. Zhong, F. Gao, T. Liu, J. Wang, J. Sun, and H. Zhou, "Scattering characteristics guided network for ISAR space target component segmentation," *IEEE Geosci. Remote Sens. Lett.*, early access, Jun. 25, 2025, doi: 10.1109/LGRS.2025.3576662.
[48] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.
[49] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, 2017.
[50] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 6467–6476.



Xiping Shang received the B.S. degree from Qingdao University of Technology, Qingdao, China, in 2017, and the M.S. degree from Harbin University of Science and Technology, Harbin, China, in 2021. He is currently pursuing the Ph.D. degree with the School of Electronic and Information Engineering, Beihang University, Beijing, China.

His research interests include multimodal learning, automatic target recognition, and computer vision.



Wei Zhao received the B.S. degree, the M.S. degree, and the Ph.D. degree from the School of Automatic Control, Northwestern Polytechnical University, Xi'an, China, in 1994, 1997, and 2000, respectively.

She was a Post-Doctoral Researcher at Beihang University, Beijing, China, where she is currently a Full Professor. Her main research interests include digital image processing, automatic target recognition, and information fusion.



Haoxiang Chen received the B.S. degree from Zhengzhou University, Zhengzhou, China, in 2021. He is currently pursuing the Ph.D. degree in signal and information processing with Beihang University, Beijing, China.

His research interests include object tracking and computer vision.



Xudong Fan received the B.S. degree in applied physics from the Ecole Centrale de Pékin, Beihang University, Beijing, China, in 2017, and the M.S. degree in computer science and engineering from the École Polytechnique, Université Libre de Bruxelles, Brussels, Belgium, in 2019. He is currently pursuing the Ph.D. degree in signal and information processing from Beihang University.

His research interests include image processing and computer vision.

Nannan Li, photograph and biography not available at the time of publication.

Dongjin Li, photograph and biography not available at the time of publication.

Jianwei Lv, photograph and biography not available at the time of publication.

Rufei Zhang, photograph and biography not available at the time of publication.

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